

Phase A Methodology: A Causal Audit of Semantic Induction Heads on Single-hop In-context Relations

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Methodology specification — September 2026

1 Goal and Guiding Principles

Ren et al. [1] define *semantic induction heads* (SIH) observationally, via the Relation Index (RI), and report correlations with ICL emergence — with no interventions. Phase A asks: **do heads passing the SIH definition causally carry the model’s single-hop in-context relation ability, and is the definition complete — or are these heads the visible tip of a larger circuit?** (Phase B extends to two-hop composition.)

Four principles constrain everything below:

1. **Audit, not verification.** We measure the observational score *and* the causal effect on *all* $32 \times 32 = 1024$ heads, then compare the two rankings. Selecting heads by RI and testing only those would make the definition’s misses invisible by construction.
2. **Definitional fidelity.** Parameters inside the SIH definition (notably the dominance threshold $\tau = 2.2$) are kept exactly as published; we may not redesign the thing we are auditing. Deviations appear only as sensitivity analyses.
3. **Null-calibrated, pre-registered thresholds.** Every selection threshold is a *rule* fixed before seeing results, calibrated against an explicit null distribution — never a percentile of the empirical score distribution (which would “select” a fixed fraction of heads even under pure noise).
4. **Discovery / validation split.** All screening and selection uses the 89 discovery families (even family ids); the 87 validation families are touched only by the final faithfulness and generalization tests.

2 Experimental Substrate

Model. OLMo-2-1124-7B (base), revision pinned in `model_lock_olmo2.json`; fp16 across two 11–12 GB GPUs. Pythia-6.9B retained as a contrast model for later comparisons.

Data. `pilot_v3/data_singlehop/singlehop_v1_*.jsonl`: 200 kinship families (relation *mother-of* over fictitious names), each with three variants $\times$ two fact orders: *base*; *corrupted* — the two answer-side names swapped inside the fact lines, so the former distractor becomes the correct answer while every mention count is unchanged and the two prompts are **token-aligned position-by-position** (≤ 6 differing tokens); *reorder* — fact lines shuffled. Every fact line (demonstrations included) carries construction-time annotations: relation, head/tail entities, and their character spans — the anchors RI needs, with no parser noise (an improvement over [1]’s spaCy/AGENDA

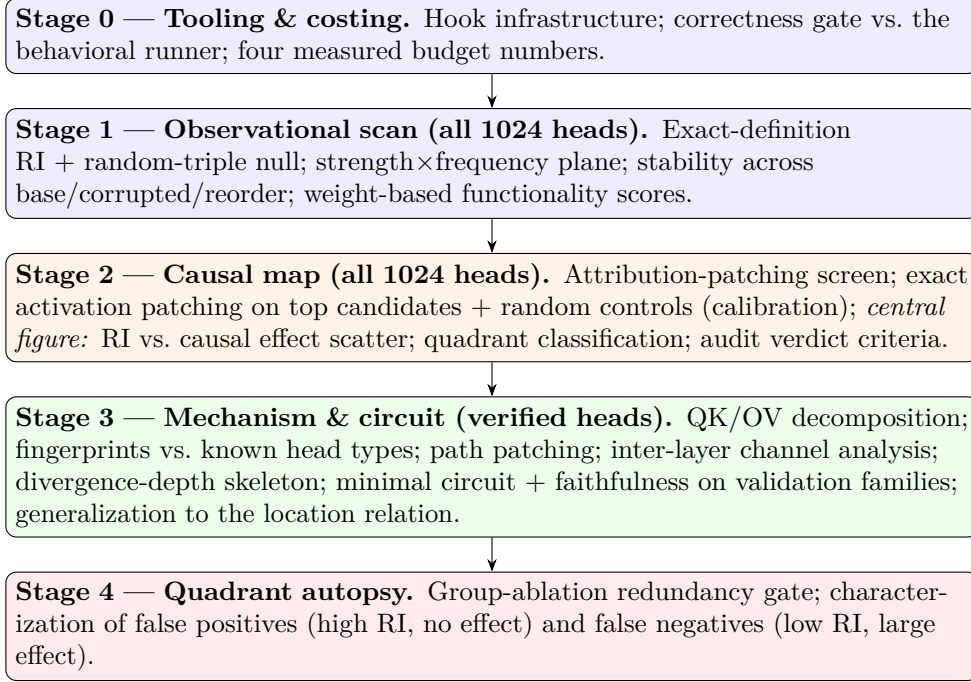


Figure 1: Phase-A pipeline. Stages 1–2 are computed on *every* head; stage 3 only on causally verified heads.

annotation). A separate 50-family *location* world (relation *inside*) is reserved for the stage-3 generalization test.

Working condition. 4-shot with single-hop-only demonstrations (behavioral baseline: base 0.93, corrupted 0.94, reorder 0.89 both-orders accuracy; working set of 176/200 families correct on base+corrupted in both orders). 0-shot is the secondary contrast. Demonstration sets are identical across all variants and orders of a family, so twins differ only in the test block.

Primary metric for all interventions. The candidate logit difference at the answer position,

$$M(x) = \text{logit}(y_{\text{gold}} \mid x) - \text{logit}(y_{\text{distractor}} \mid x), \quad (1)$$

following the recommendations of [6] (logit difference rather than probability or accuracy: linear in the final residual stream, sensitive, sign-interpretable). On a clean/corrupted twin the gold and distractor swap roles, so patching effects have a *signed* prediction (the difference should flip toward the corrupted run’s answer), not merely “performance drops”.

3 Stage 0 — Tooling and Costing

What. Stand up hook-based access to internal activations (read attention patterns and per-head outputs; write/replace head outputs mid-run) via TransformerLens [14], with raw PyTorch forward hooks on the HF model as fallback if the wrapper misbehaves on our sharded 2-GPU setup.

Correctness gate. Before any measurement: the hooked model must reproduce the behavioral runner’s candidate scores on 20 prompts within numerical tolerance. No gate, no science.

Why. All later budgets depend on four numbers that cannot be reliably guessed for this hardware.

#	Measurement	What it budgets
1	sec + GiB of a forward pass with attention caching	Stage-1 scan time ($\times$ #prompts); cache is $L H n^2 \approx 32 \cdot 32 \cdot 350^2 \approx 1.3 \times 10^8$ values/prompt $\Rightarrow$ streaming policy
2	sec of a full RI computation per prompt	Verifies the cheap order of operations (QK filter from cached attention first; OV vocabulary projection only for surviving (head, position) pairs at entity positions — naively this stage costs ~ 30 forwards/prompt)
3	sec of one forward with a hook intervention	Unit price of exact patching (stage-2 verification budget)
4	does a backward pass fit in GPU memory?	Gates attribution patching (needs $\partial M / \partial a$ for all sites); if not: 4-GPU jobs or memory-saving fallbacks

4 Stage 1 — Observational Scan (all heads)

4.1 Relation Index, exact definition

Per head h , per annotated triple with source-entity anchor token t_s (position s) and target-entity anchor t_o (position o), evaluated at positions $j \geq \max(s, o)$ in the prompt:

1. **QK stage (selection)**. The head must attend from j to s dominantly:

$$s = \arg \max_k A_{j,k}^h \quad \text{and} \quad A_{j,s}^h / \max_{k \neq s} A_{j,k}^h > \tau, \quad \tau = 2.2. \quad (2)$$

τ is kept at the published value (principle 2). Because [1] chose 2.2 as the ≈ 95 th percentile of the dominance-ratio distribution *in their setting*, we also recompute that recipe on our model and data, obtaining τ_{ours}^* : if $\tau_{\text{ours}} \approx 2.2$ the constant transfers and we report that; if not, the scan is run under both values and the discrepancy is itself an audit finding (the definition does not transfer stably across models). Sensitivity is thus reported at $\tau \in \{2.2, \tau_{\text{ours}}^*, 3.0\}$.

2. **OV stage (promotion)**. Project the attended representation through the head to the vocabulary,

$$p^{h,j} = \text{softmax}(x W_{OV}^h W_U), \quad q_t^{h,j} = \max(0, p_t^{h,j} - \mathbb{E}_{t'}[p_{t'}^{h,j}]), \quad (3)$$

and score the target’s share among context tokens: $a^{h,j} = q_{t_o}^{h,j} / \sum_t q_t^{h,j}$.

Multi-token entities: the anchor is the *last* token of the entity span (identity complete there); robustness to the first-token convention is checked once on a subsample. Anchors come from the datasets’ **facts** field. Triples from demonstration blocks and the test block are scored together (more statistics per forward), reported also separately.

4.2 Two head-level statistics and their null

For each head we report a point on a plane, not a single number:

$$\text{frequency}(h) = \Pr[\text{QK stage passes}], \quad \text{strength}(h) = \mathbb{E}[a^{h,j} \mid \text{QK passed}]. \quad (4)$$

Null distribution (the “random-triple floor”). The identical computation with t_o replaced by a random context token (re-drawn per triple), following [1]’s fake-target control. **Pre-registered selection rule**: a head is an *RI-passing head* if its strength exceeds the 99.9th percentile of the

null (≈ 1 expected false positive among 1024 heads; with two statistics tested at the “pass either” rule the expectation is ≈ 2); sensitivity reported at the 99th and 99.5th percentiles. Percentiles of the *empirical* distribution are never used for selection. The empirical distribution itself (separated tail? bimodality?) is reported as a finding.

4.3 Stability across variants and the attention-side signed test

The scan runs on base, corrupted, and reorder prompts of the discovery families. For each head: does the QK stage *track the swap* in corrupted prompts (attend to the new source), and does it survive reorder? Heads whose RI collapses under reorder are layout-driven by the same logic as the behavioral order-pair design.

4.4 Weight-based functionality scores (all heads, offline)

Following the parameter-space analysis framework of [12] and the projections of [3], we compute for every head, from weights alone (no forward passes): (i) a *copying-structure* score: dominance and sign of the diagonal of $W_U W_{OV}^h W_E$ restricted to name tokens (the spectral copying diagnostic of [3]); (ii) a *token-matching* score: diagonal dominance of $W_E^\top W_{QK}^h W_E$ on name tokens; (iii) a *relation-word matching* score: the QK projection mass between occurrences of the relation word (“mother”). **Why:** our relation is in-context over fictitious names, so no parametric relation lives in the weights; the structural prediction for an in-context relation head is *search-and-copy machinery*. These scores are a third, method-independent evidence axis, and the primary tool of stage 4’s autopsy. Exact normalization recipes are taken from [12] at implementation time.

5 Stage 2 — Full Causal Map (all heads)

5.1 Attribution-patching screen

Exact activation patching for all heads is infeasible (1024×178 twin pairs $\approx 1.8 \times 10^5$ forwards). Instead, a first-order estimate of every head’s patching effect is obtained from three passes per twin pair [7]:

$$\widehat{\Delta M}_h = (a_h^{\text{corr}} - a_h^{\text{clean}})^\top \left. \frac{\partial M}{\partial a_h} \right|_{\text{clean}}, \quad (5)$$

where $a_h \in \mathbb{R}^{d_{\text{model}}}$ is head h ’s output at the answer position. One clean forward, one corrupted forward, and one backward pass of M yield $\widehat{\Delta M}_h$ for *all* heads (and MLP layers, and all positions) simultaneously: ≈ 700 forward-equivalents for the full discovery set.

Known failure mode and mitigation. The estimate is a first-order Taylor approximation; it under-estimates precisely where the path from the activation to the metric crosses saturated softmax plateaus — i.e., it is biased *against* sharply-attending heads, our most interesting candidates. Therefore:

1. **Calibration (pre-registered):** exact activation patching on the union of the top-25 heads by each ranking (RI, attribution, weight scores) plus 25 random heads, on 40 twin pairs. The exact-vs-estimate agreement (Spearman correlation) on this sample is itself a reported result; if $\rho < 0.7$, the screen is upgraded to integrated-gradients attribution (EAP-IG, $m=5$ interpolation steps [8]) and re-calibrated.
2. Exact patching is always the measurement of record for any head individually discussed.

5.2 The central figure and the audit verdict

The main deliverable: a scatter of all 1024 heads, x = RI strength, y = causal effect (calibrated estimate; exact where available), color = weight-based copy score. Threshold-free statistics reported: Spearman rank correlation between RI and causal effect over all heads; overlap curves (top- k Jaccard).

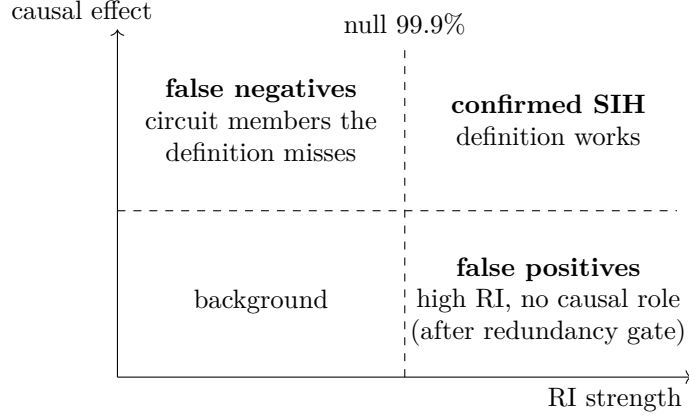


Figure 2: Quadrant reading of the central scatter. Stage 4 characterizes the two off-diagonal quadrants.

Pre-registered verdict criteria (to be confirmed with the team before stage 2 runs): the definition is judged *sound* if RI-passing heads have systematically larger causal effects than layer-matched non-passing heads; *incomplete* if at least one head in the top decile of exact-patching effect lies below even the *lenient* (99th-percentile) RI null threshold — so that the incompleteness claim cannot be manufactured by our own strictness; *misleading* if the median causal effect of RI-passing heads does not exceed that of random heads. These are not mutually exclusive (e.g., sound and incomplete). Note that the selection threshold never gates *which heads are studied* — the stage-2 causal map covers all 1024 heads unconditionally; the threshold only names the set attributed to the definition, and the primary quantitative claims are threshold-free (rank correlation, overlap curves).

6 Stage 3 — Mechanism and Circuit (verified heads only)

Tools, in order of application:

1. **QK side — edge-following score.** Using the fact anchors: attention mass from the answer position (and from the question-entity mention) to the correct fact’s tail token, vs. positional controls; *signed* version on corrupted twins (does attention move to the swapped source?).
2. **OV side — what is written.** Project the head’s actual output at the answer position to the vocabulary; verify it promotes the specific answer name; spectral copying check on the name subspace [3].
3. **Fingerprints vs. known head types.** Standardized scores locate our heads in the existing taxonomy, answering “new kind, or re-labeled known kind?”: prefix-matching and copying scores on random repeated sequences [2]; retrieval-head score (fraction of copy events where the head’s top attention sits on the copied token) [10]; function-vector mediation (does the head’s mean output over relation prompts causally induce the relation zero-shot?) [11]. Deliverable: a head $\times$ fingerprint table.

4. **Path patching** [4, 5]: which upstream components feed the verified heads’ queries/keys/values, and which downstream components consume their output — the head-vs-circuit question at edge resolution. Position-aware edges follow [9] given our templated prompts.
5. **Inter-layer channel content.** For each strong edge $A \rightarrow B$ found by path patching, SVD of the writer-reader interaction $W_{OV}^A W_{QK}^B$ to identify the low-rank communication subspace and interpret the message [13] — the content-level refinement of [3]’s composition-strength norm.
6. **Divergence-depth skeleton.** Per layer ℓ , the residual-stream distance at the answer position between the token-aligned twins, $d(\ell) = \|h_\ell^{\text{clean}} - h_\ell^{\text{corr}}\|$. The layer where d jumps is where the swap’s consequence reaches the answer position; all head-level findings must be consistent with this curve (a “critical” head above the completed divergence is a contradiction to explain). Nearly free from cached runs; comparison point to the layer-level analyses of parametric multi-hop work.
7. **Minimal circuit and faithfulness.** Define the circuit as the verified nodes + strong edges; run with everything else mean-ablated (means over discovery prompts); **pre-registered criterion:** the circuit must recover $\geq 80\%$ of the full model’s clean–corrupted logit-difference gap on the *validation* families. Completeness check: ablating the circuit from the full model must destroy the behavior.
8. **Generalization.** Repeat the RI scan + exact patching of the identified heads on the *location* world: same heads $\Rightarrow$ general in-context relation machinery; disjoint heads $\Rightarrow$ relation-specific — either outcome is a result.

7 Stage 4 — Quadrant Autopsy

Gate first: group ablations. A single-head zero effect does not prove irrelevance — IOI-style backup heads activate when a primary head is removed [4]. For each false-positive candidate: ablate jointly with all similarly-profiled heads and measure effect vs. coalition size. Only heads inert *in coalition* are declared causally irrelevant; the effect-vs-coalition-size curve doubles as a quantitative distributedness measure.

False positives (high RI, no causal role). What are they instead? Weight fingerprints (stage 1), attention anatomy on our prompts (position? punctuation? relation word? recency?), and behavior on generic copying sequences. Deliverable: an alternative explanation for each head’s high RI — the empirical content behind “the definition over-triggers”.

False negatives (low RI, large effect). What does the definition miss? Expected from our direct-path-blindness analysis: heads whose contribution is indirect (e.g., moving information between fact tokens rather than to the answer position), invisible to a metric that only reads the direct path to the logits. Characterized with the same tools; this quadrant, if populated, is the audit’s sharpest finding.

Afterwards, as time permits: Phase B (two-hop composition, using the same verified single-hop circuit as the component inventory) and/or the checkpoint sweep.

8 Summary of Pre-registered Decisions

Item	Rule
SIH definition parameters	$\tau = 2.2$ exactly; sensitivity $\{1.5, 3.0\}$ in appendix
Head selection threshold	99.9th percentile of the random-triple null (per statistic); sensitivity at 99/99.5
Screen quality gate	Spearman(exact, attribution) ≥ 0.7 on the calibration sample, else upgrade to EAP-IG
Faithfulness criterion	$\geq 80\%$ recovery of the clean-corrupted logit-diff gap on validation families
Audit verdict definitions	sound / incomplete / misleading, as specified in §5.2
Selection data	discovery families only; validation families reserved for faithfulness + generalization

References

- [1] J. Ren et al. (2024). *Identifying Semantic Induction Heads to Understand In-Context Learning*. Findings of ACL. <https://aclanthology.org/2024.findings-acl.412/>
- [2] C. Olsson et al. (2022). *In-context Learning and Induction Heads*. Transformer Circuits Thread. <https://transformer-circuits.pub/2022/in-context-learning-and-induction-heads/index.html>
- [3] N. Elhage et al. (2021). *A Mathematical Framework for Transformer Circuits*. Transformer Circuits Thread. <https://transformer-circuits.pub/2021/framework/index.html>
- [4] K. Wang et al. (2022). *Interpretability in the Wild: a Circuit for Indirect Object Identification in GPT-2 Small*. <https://arxiv.org/abs/2211.00593>
- [5] S. Heimersheim and N. Nanda (2024). *How to Use and Interpret Activation Patching*. <https://arxiv.org/abs/2404.15255>
- [6] F. Zhang and N. Nanda (2024). *Towards Best Practices of Activation Patching in Language Models: Metrics and Methods*. ICLR. <https://arxiv.org/abs/2309.16042>
- [7] A. Syed, C. Rager, A. Conmy (2023). *Attribution Patching Outperforms Automated Circuit Discovery*. <https://arxiv.org/abs/2310.10348>
- [8] M. Hanna, S. Pezzelle, Y. Belinkov (2024). *Have Faith in Faithfulness: Going Beyond Circuit Overlap When Finding Model Mechanisms*. <https://arxiv.org/abs/2403.17806>
- [9] T. Haklay et al. (2025). *Position-aware Automatic Circuit Discovery*. <https://arxiv.org/abs/2502.04577>
- [10] W. Wu et al. (2024). *Retrieval Head Mechanistically Explains Long-Context Factuality*. <https://arxiv.org/abs/2404.15574>
- [11] E. Todd et al. (2024). *Function Vectors in Large Language Models*. ICLR. <https://arxiv.org/abs/2310.15213>
- [12] A. Elhelo and M. Geva (2024). *Inferring Functionality of Attention Heads from their Parameters*. <https://arxiv.org/abs/2412.11965>
- [13] J. Merullo, C. Eickhoff, E. Pavlick (2024). *Talking Heads: Understanding Inter-layer Communication in Transformer Language Models*. <https://arxiv.org/abs/2406.09519>

- [14] N. Nanda and J. Bloom. *TransformerLens*. <https://github.com/TransformerLensOrg/TransformerLens>